

Isotopie

CHM-4152
Chimie de l'environnement

Raoul Couture
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raoul.couture@chm.ulaval.ca



Nomenclature: structure atomique



X = Symbole élémentaire
A = Nombre de masse
Z = Numéro atomique



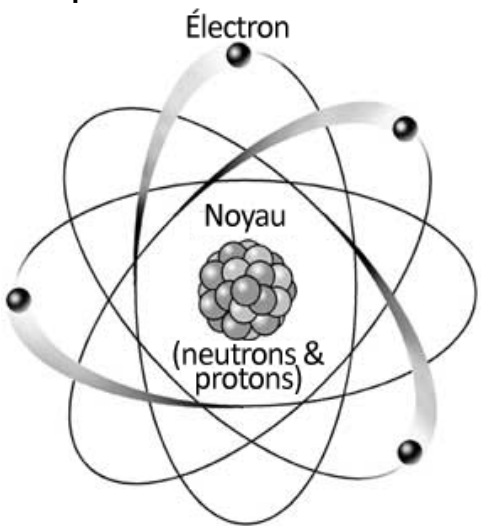
$m_{\text{électron}} = 9.1085 \times 10^{-31} \text{ kg}$

$m_{\text{proton}} = 1.6726 \times 10^{-27} \text{ kg}$

$m_{\text{neutron}} = 1.67492 \times 10^{-27} \text{ kg}$

$A = Z + N$

où Z et N sont le nombre de protons et neutrons, respectivement.



Isotopes stables vs radioisotopes

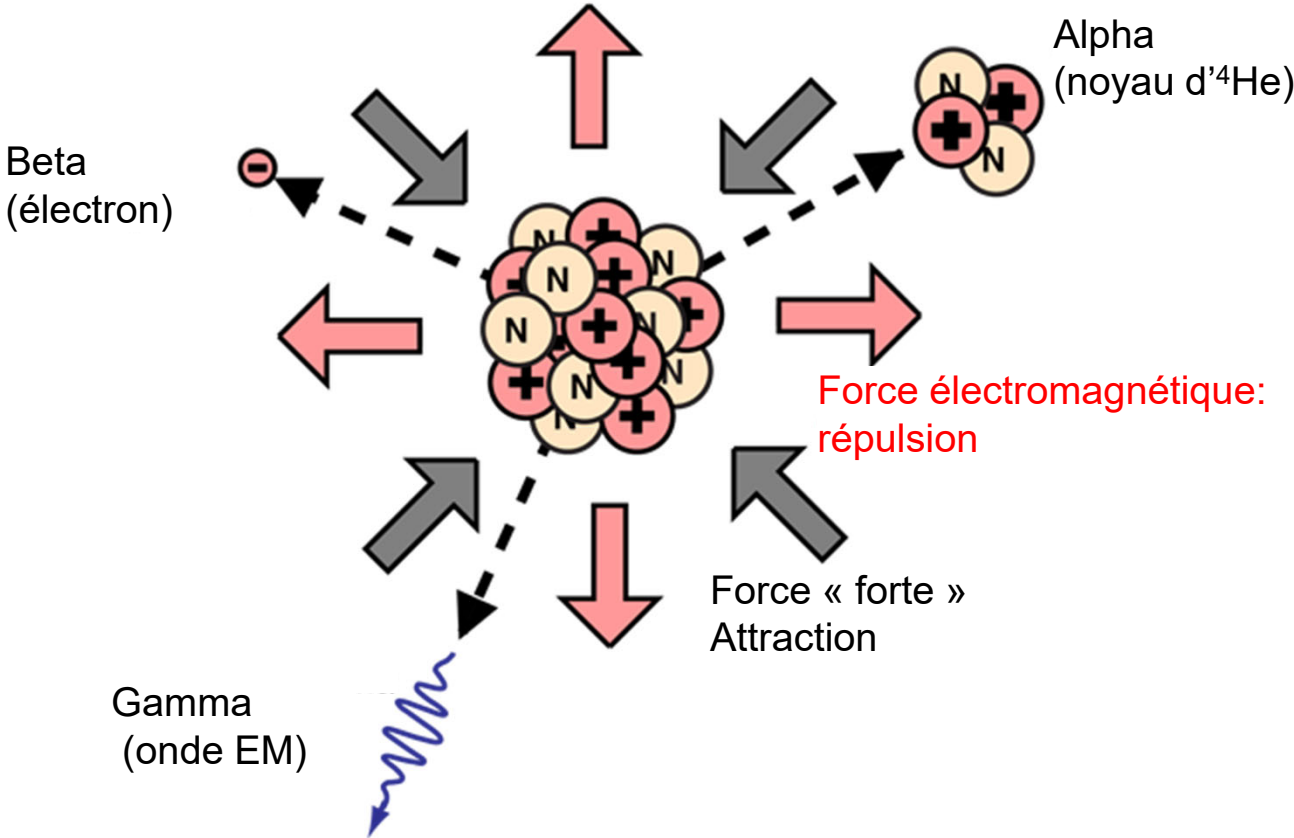
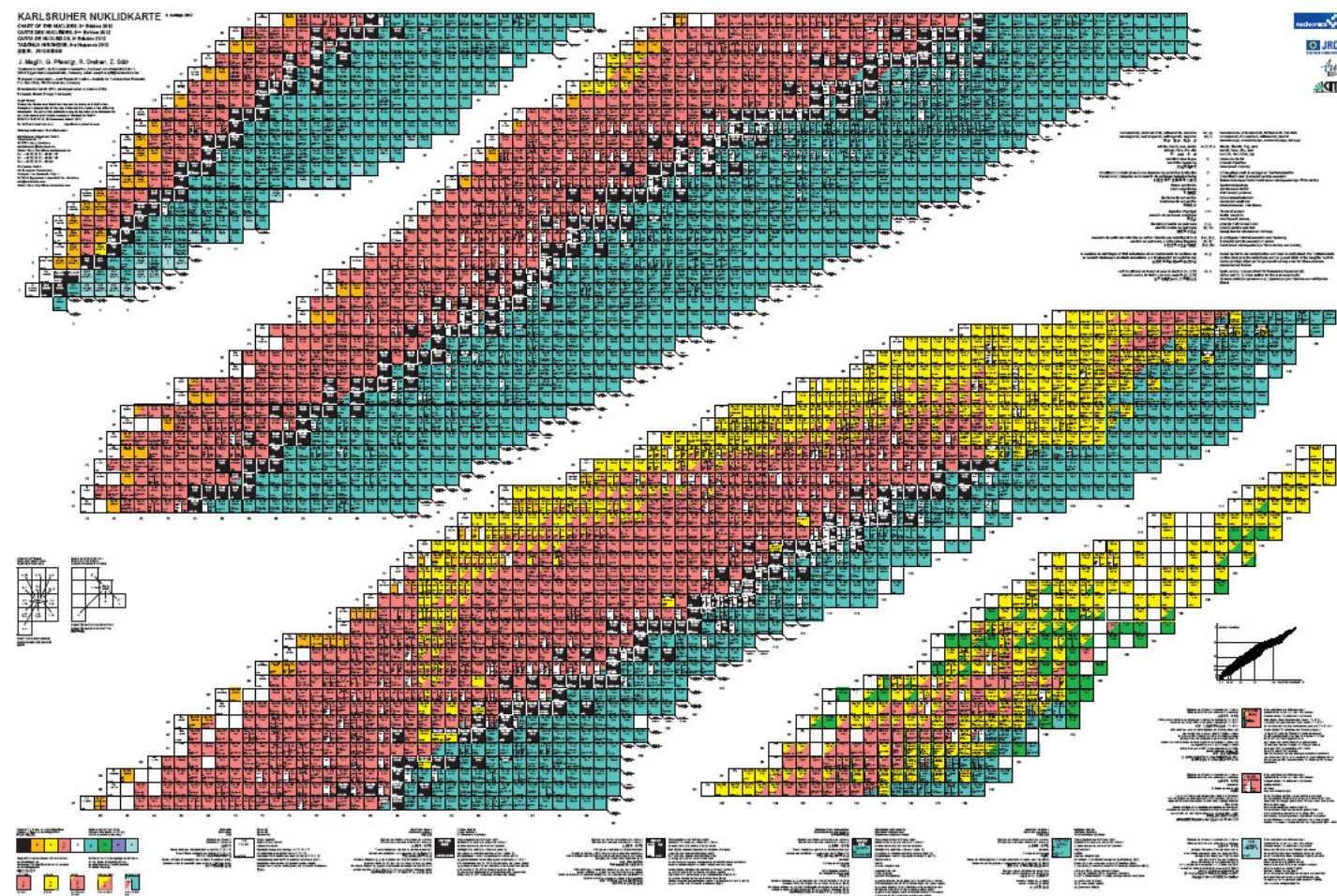


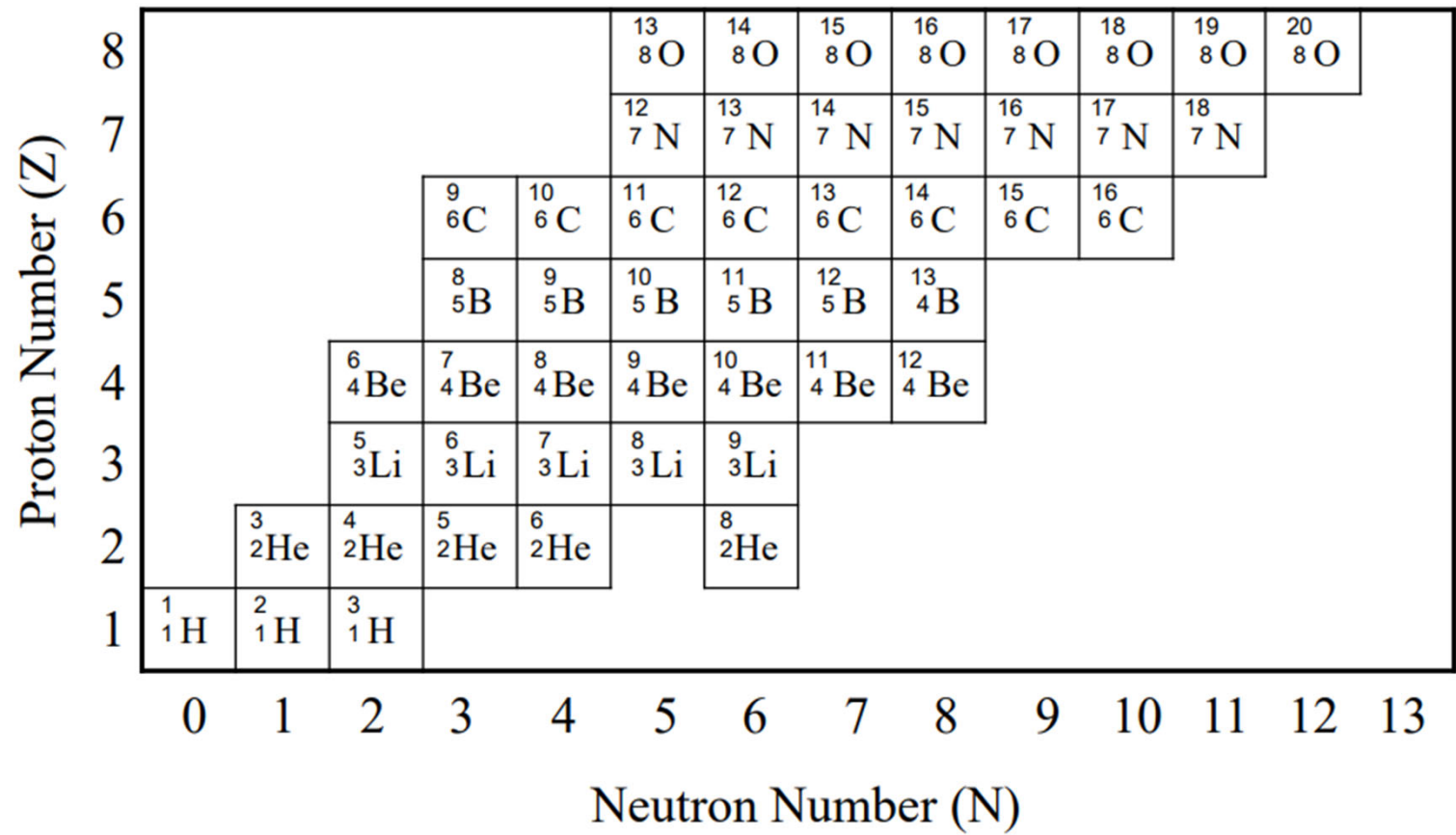
Image adaptée de Rod Nave's hyperphysics : hyperphysics.phy-astr-gsu.edu/hbase/Nuclear/nucnot.html

Carte des nucléides: isotopes (protons)



Karlsruher Nuclid Chart 12th edition.

Carte des nucléides: isotopes (protons)

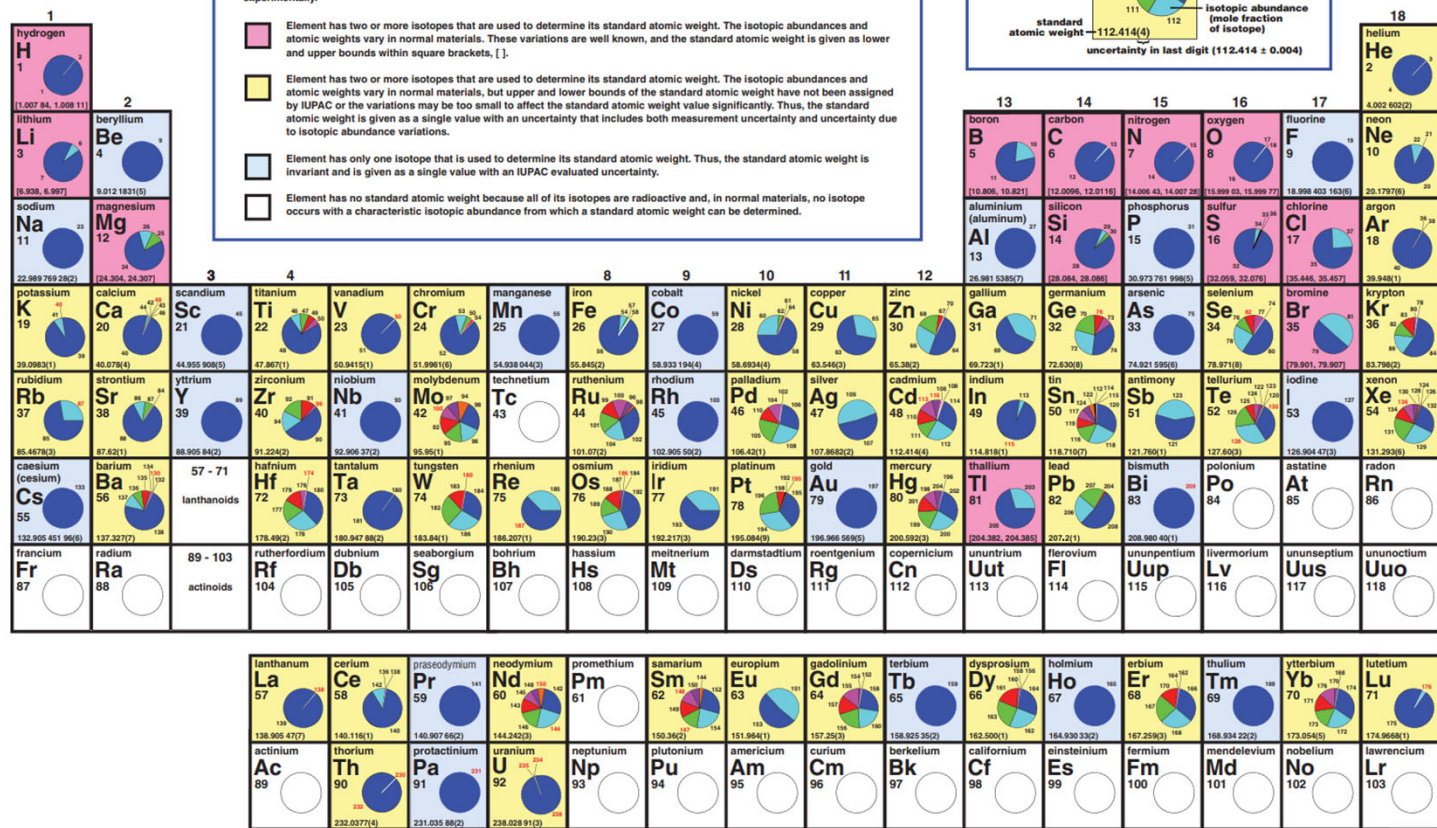
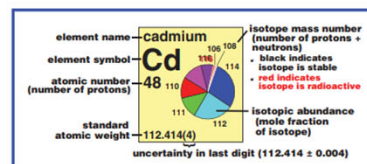


IUPAC Periodic Table of the Isotopes

Element Background Color Key

Standard atomic weights are the best estimates by IUPAC of atomic weights that are found in normal materials, which are terrestrial materials that are reasonably possible sources for elements and their compounds in commerce, industry, or science. They are determined using all stable isotopes and selected radioactive isotopes (having relatively long half-lives and characteristic terrestrial isotopic compositions). Isotopes are considered stable (non-radioactive) if evidence for radioactive decay has not been detected experimentally.

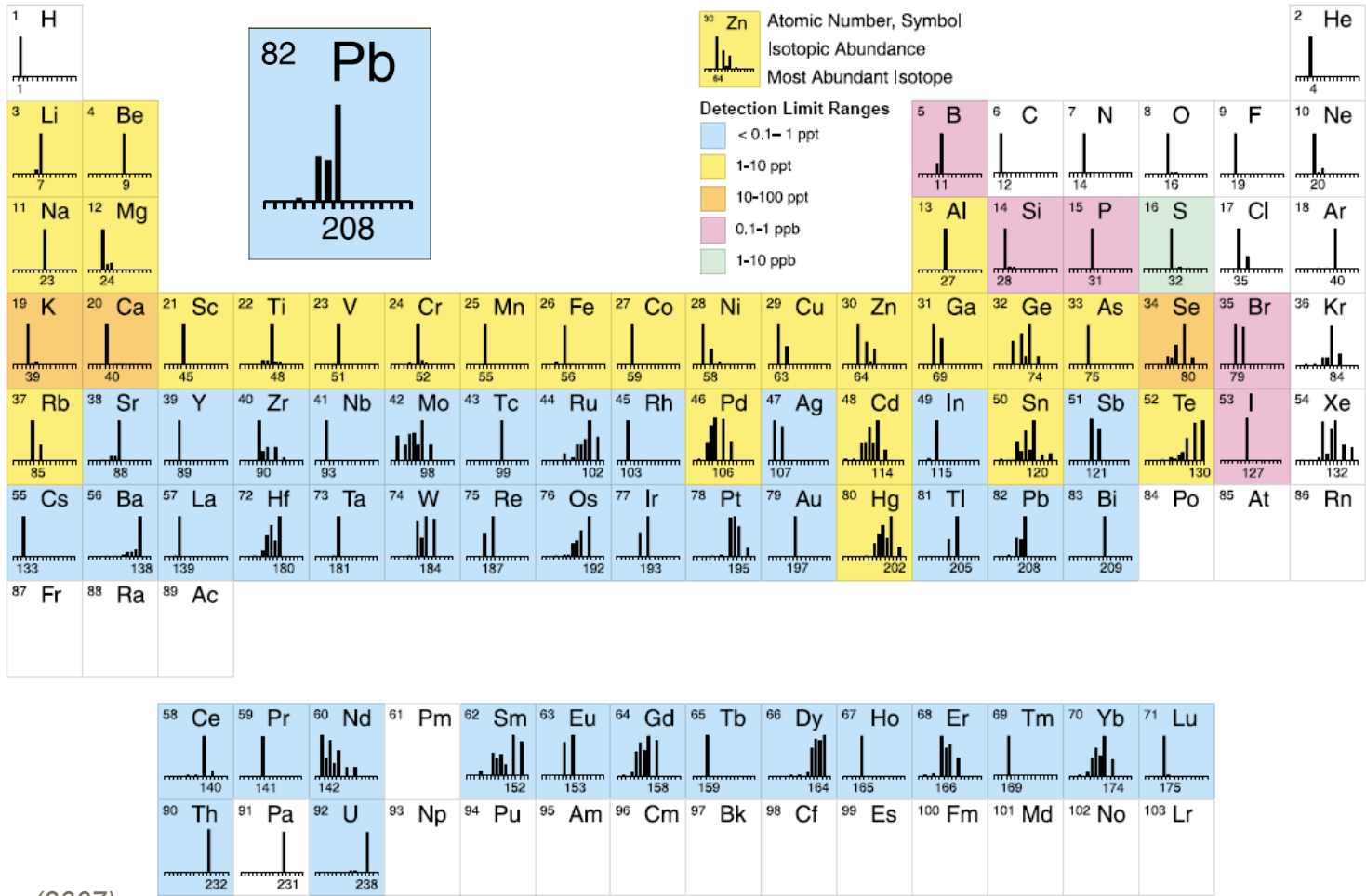
-  Element has two or more isotopes that are used to determine its standard atomic weight. The isotopic abundances and atomic weights vary in normal materials. These variations are well known, and the standard atomic weight is given as lower and upper bounds within square brackets, [].
-  Element has two or more isotopes that are used to determine its standard atomic weight. The isotopic abundances and atomic weights vary in normal materials, but upper and lower bounds of the standard atomic weight have not been assigned by IUPAC or the variations may be too small to affect the standard atomic weight value significantly. Thus, the standard atomic weight is given as a single value with an uncertainty that includes both measurement uncertainty and uncertainty due to isotopic abundance variations.
-  Element has only one isotope that is used to determine its standard atomic weight. Thus, the standard atomic weight is invariant and is given as a single value with an IUPAC evaluated uncertainty.
-  Element has no standard atomic weight because all of its isotopes are radioactive and, in normal materials, no isotope occurs with a characteristic isotopic abundance from which a standard atomic weight can be determined.



International Union of Pure and Applied Chemistry (IUPAC) Project 2007-038-3-200, "Development of an isotopic periodic table for the educational community"

October 1, 2013 www.iaaw.org

Tableau périodique et isotopes stables



PerkinElmer. (2007).

Pourquoi mesurer les isotopes ?

Datation :

les isotopes issus de la désintégration d'élément radioactif

Source de contaminants:

origine, transfert et mélange de minéral

Paléothermomètres:

températures de l'océan, de l'atmosphère, de cristallisation

Géochimie:

mécanismes responsables de la mobilité des éléments chimiques

Notation: ratio, delta et facteur de fractionnement

Composition isotopique

$$R_A = \frac{\text{quantité de H}_2^{18}\text{O}}{\text{quantité de H}_2^{16}\text{O}}$$

Facteur de fractionnement

Normalement, l'isotope le plus lourd est utilisé au numérateur

$$\alpha_{A-B} = \frac{R_A}{R_B}$$

$$\delta^{18}\text{O} = \frac{R_A - R_{\text{ref}}}{R_{\text{ref}}} \cdot 1000 = \text{‰}$$

Expression en
« pour mille »

Composition isotopique dans les matériaux certifiés

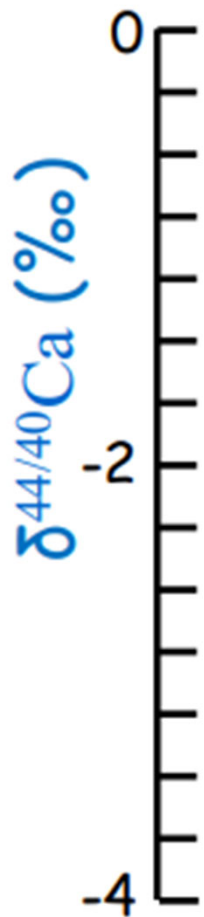
Éléments	Isotope	Abondance (%)	Différence relative de masse (%)	Standards internationaux	Abondance absolue dans standard (R_{standard})
Hydrogène	^1H	99.985	100	Vienna Standard Mean Ocean Water (VSMOW)	$^2\text{H}:^1\text{H} = 0.00015576$
	^2H (ou D)	0.0155			
Carbone	^{12}C	98.892	8.3	Vienne Pee Dee Belemnite (VPDB)	$^{13}\text{C}:^{12}\text{C} = 0.0112372$
	^{13}C	1.108			
Azote	^{14}N	99.635	7.1	Atmospheric nitrogen	$^{15}\text{N}:^{14}\text{N} = 0.0036765$
	^{15}N	0.365			
Oxygène	^{16}O	99.759	12.5 ($^{18}\text{O}:^{16}\text{O}$)	VSMOW (H ₂ O); VPDB (CO ₂ ou carbonate)	VSMOW = 0.0020052 VPDB = 0.0020672 ($^{18}\text{O}:^{16}\text{O}$)
	^{17}O	0.037			
	^{18}O	0.204			
Soufre	^{32}S	95	6.3 ($^{34}\text{S}:^{32}\text{S}$)	Vienna Canon Diablo meteriote troilite (VCDT)	$^{34}\text{S}:^{32}\text{S} = 0.0450045$
	^{33}S	0.75			
	^{34}S	4.21			
	^{36}S	0.014			

Notation: ratio, delta et facteur de fractionnement

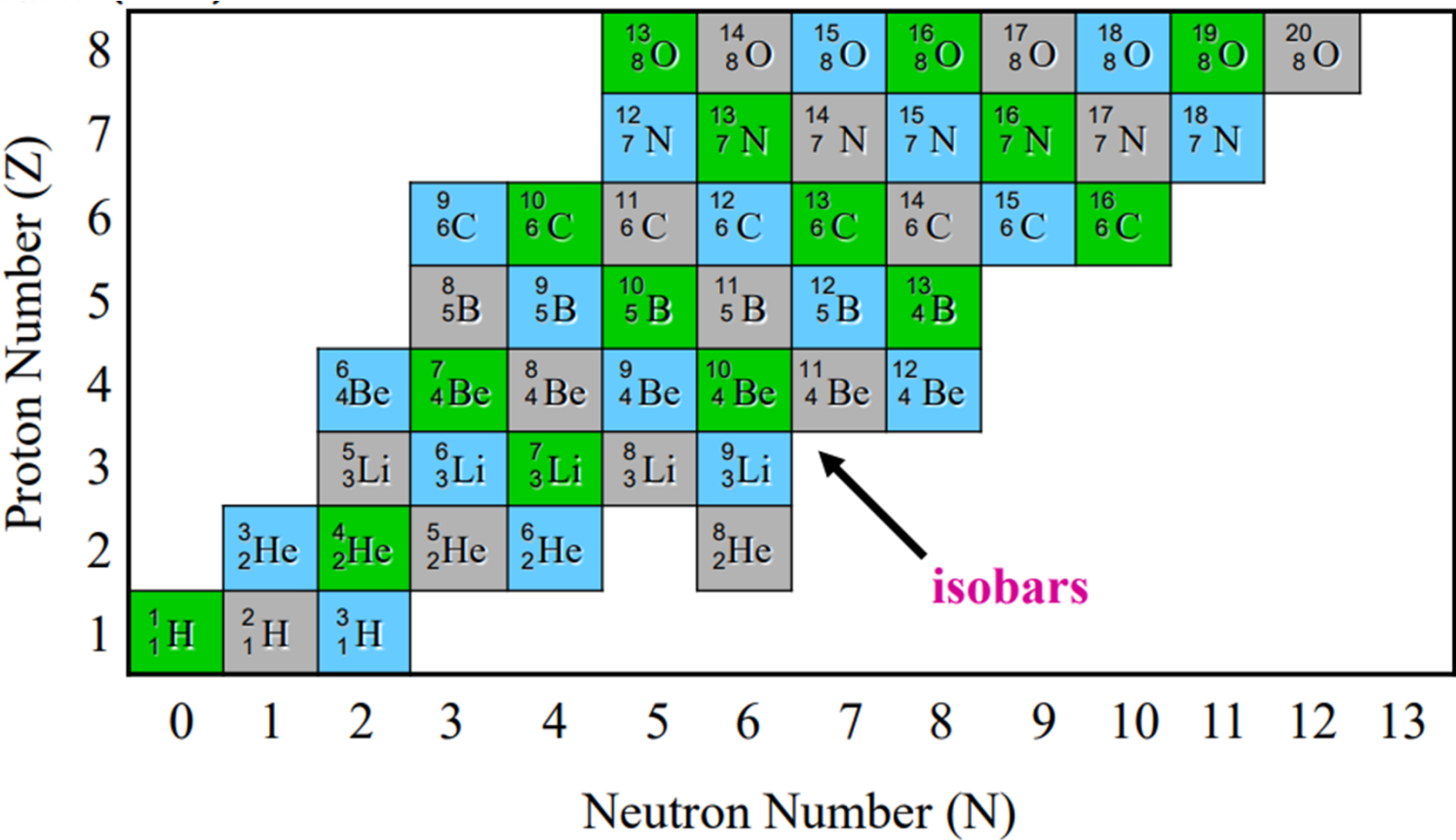
$$\delta^{44}\text{Ca} = 0 \text{ ‰}$$

$$\delta^{44}\text{Ca} > 0 \text{ ‰}$$

$$\delta^{44}\text{Ca} < 0 \text{ ‰}$$



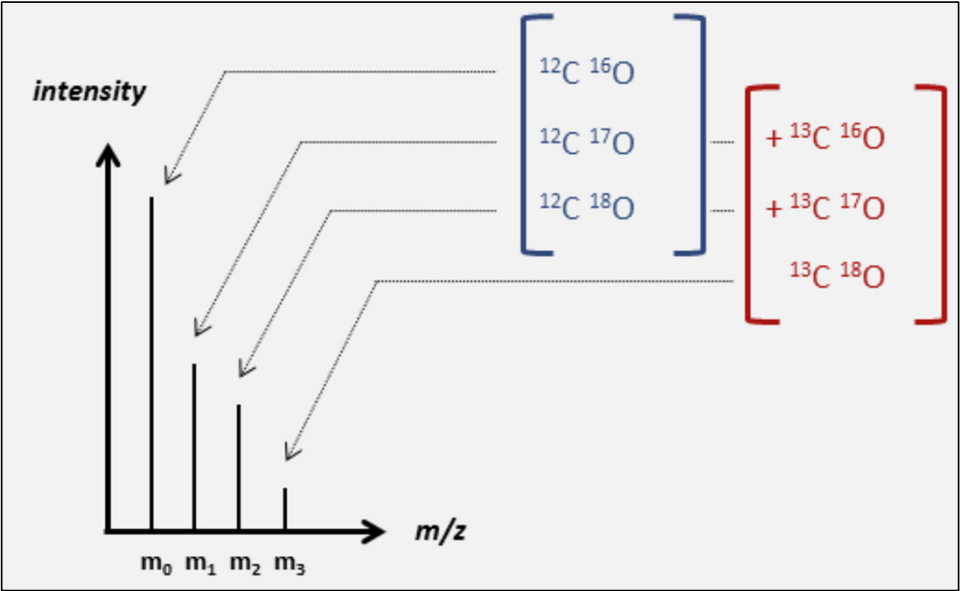
Interférences isobariques



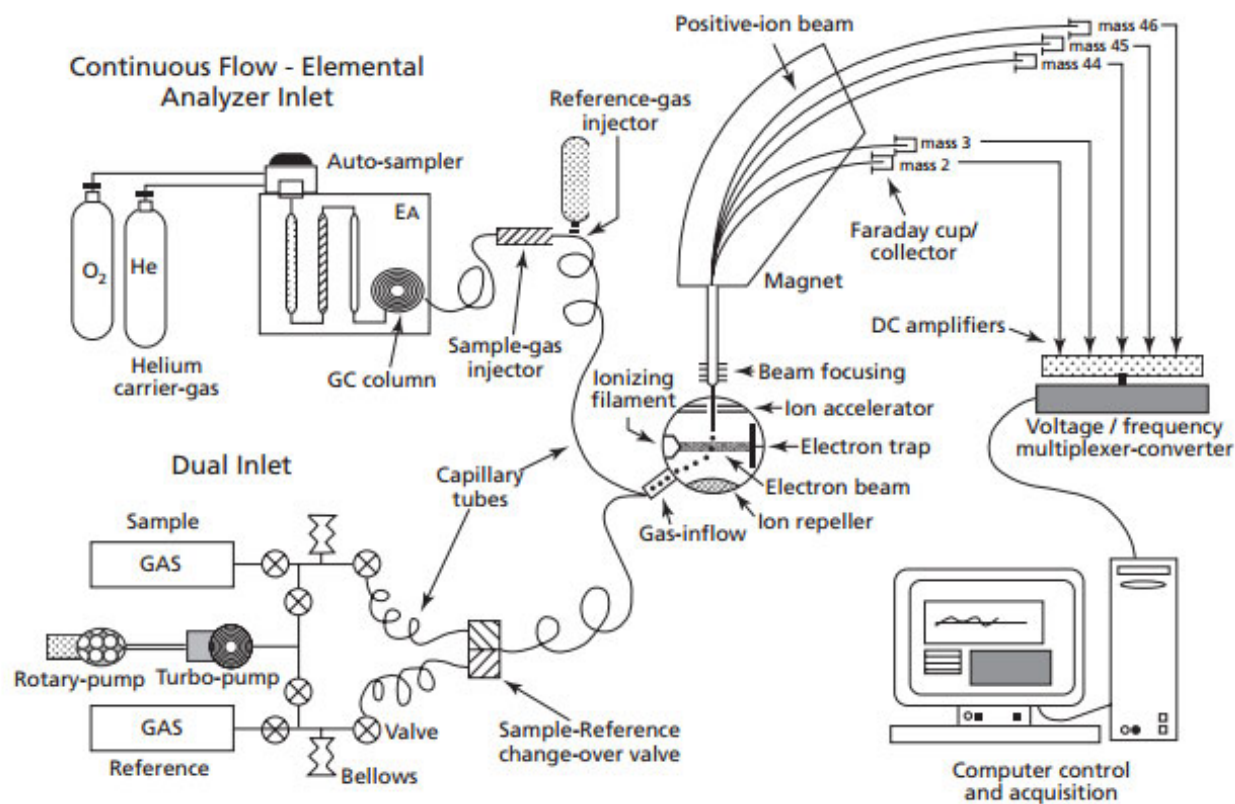
Isotopologues

Table 8.1 Relative abundances of the isotopologues of CO₂

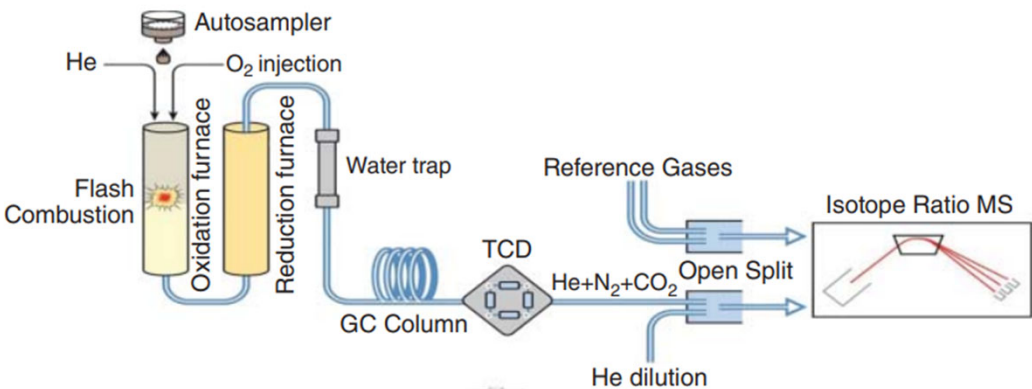
Mass	Isotopologue	Relative Abundance
44	¹² C ¹⁶ O ₂	98.40%
45	¹³ C ¹⁶ O ₂	1.11%
	¹² C ¹⁷ O ¹⁶ O	748 ppm
46	¹² C ¹⁸ O ¹⁶ O	0.40%
	¹³ C ¹⁷ O ¹⁶ O	8.4 ppm
	¹² C ¹⁷ O ₂	0.142 ppm
47	¹³ C ¹⁸ O ¹⁶ O	44.4 ppm
	¹² C ¹⁷ O ¹⁸ O	1.50 ppm
	¹³ C ¹⁷ O ₂	1.60 ppb
48	¹² C ¹⁸ O ₂	3.96 ppm
	¹³ C ¹⁷ O ¹⁸ O	16.8 ppb
49	¹³ C ¹⁸ O ₂	44.5 ppb



Dual inlet/continuous flow elemental analyzer



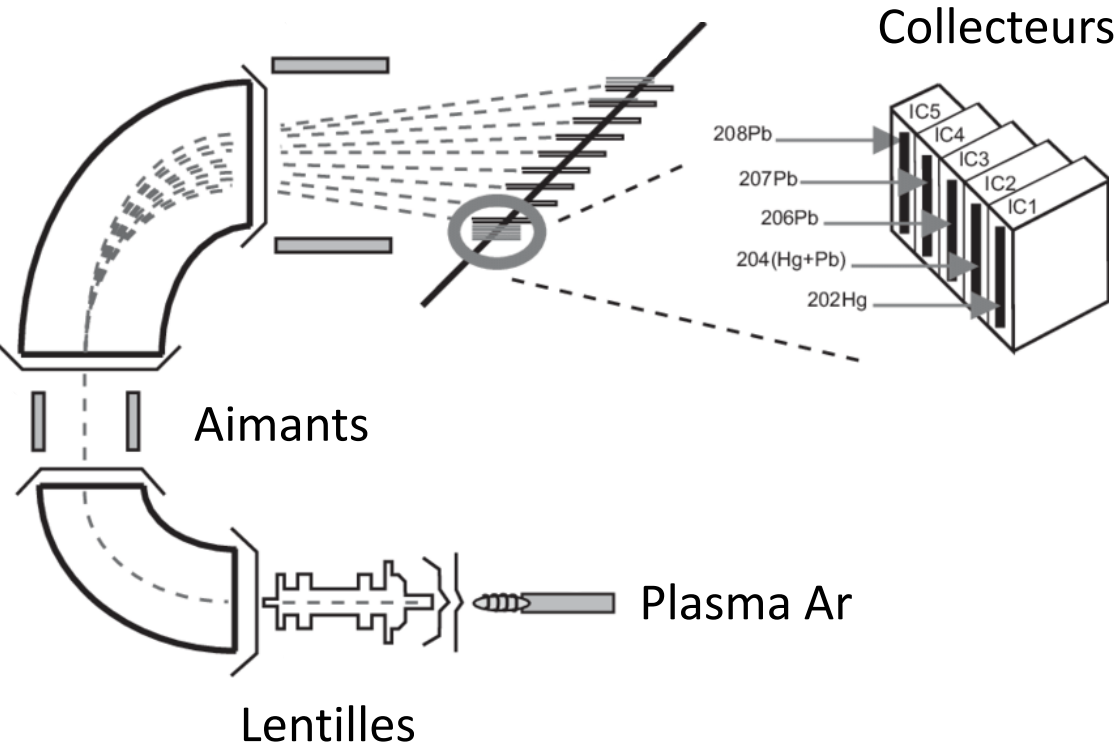
Continuous flow isotope-ratio mass spectrometer



Karapurkar et al. (2008); Thermofisher

Multi-collector ICP mass spectrometer

Thermo Neptune MC-ICP-MS



Composition isotopique

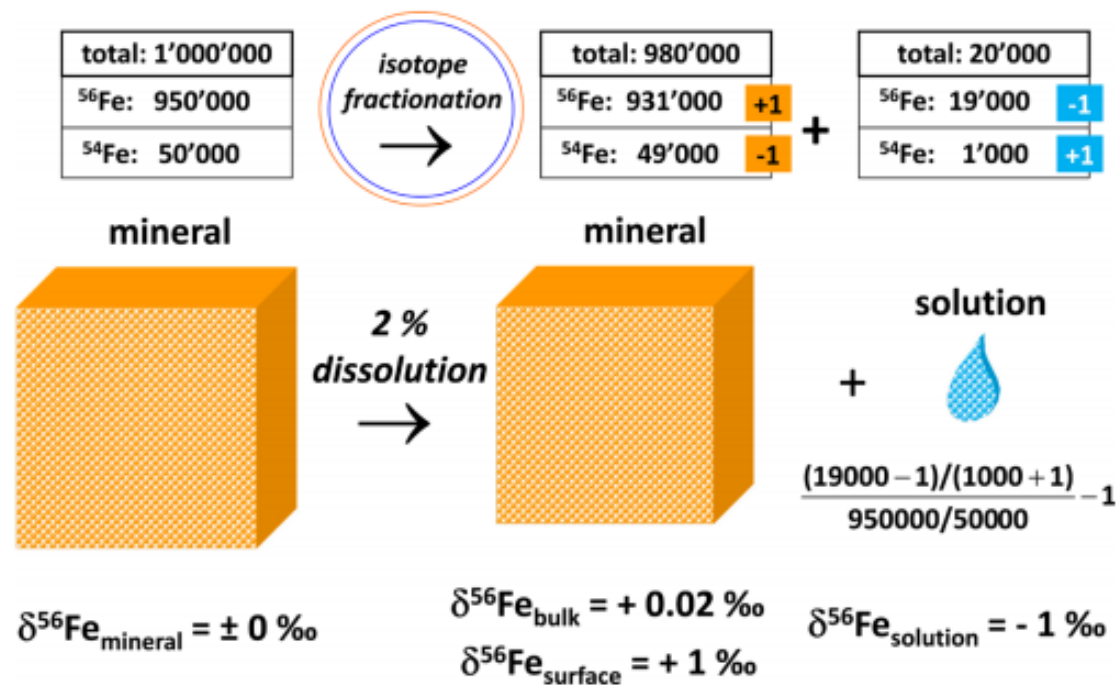
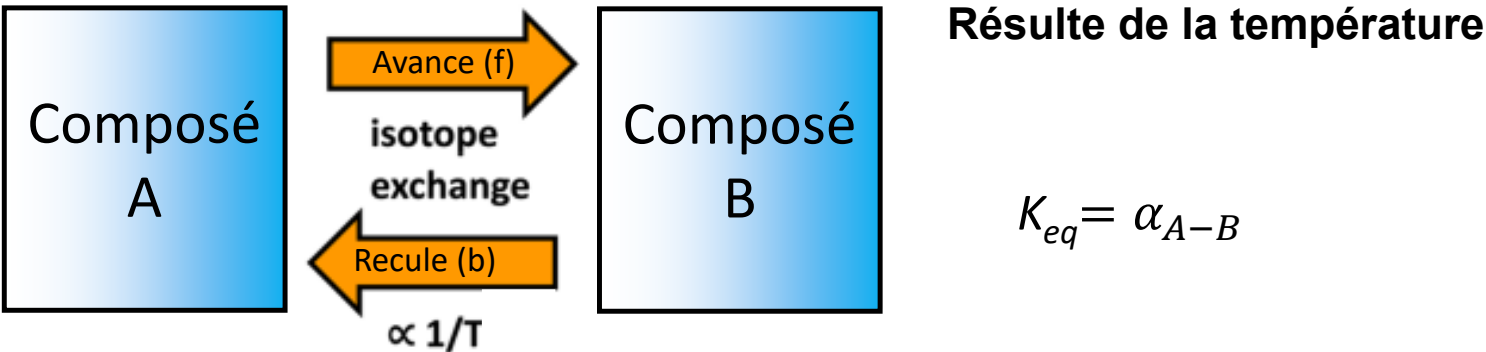


Figure 2. Schematic illustration of metal stable isotope fractionation during mineral dissolution and delta values of the involved metal pools (adapted from Wiederhold²⁹⁷). In the presented example, a small fraction (2%) of an iron mineral with 1 million Fe atoms is dissolved (simplified boundary conditions: 95% ⁵⁶Fe + 5% ⁵⁴Fe; the surface layer corresponds to 4% of the total mineral). Without isotope fractionation, the dissolved pool would contain 1000 ⁵⁴Fe atoms. However, metal isotope fractionation manifests itself by the preferential release of one light ⁵⁴Fe atom resulting in a $\delta^{56}\text{Fe}$ value of -1 ‰ in solution. The influence on the isotopic signature of the bulk mineral is negligible (+0.02 ‰) due to its larger reservoir size. However, a relative enrichment of heavy isotopes is created at the mineral surface. The general magnitude of metal stable isotope fractionation illustrated in this example is similar to the extent of fractionation observed in nature. However, in order to compare the dimensions, one has to keep in mind that one gram of soil usually contains about 10²⁰ Fe atoms.

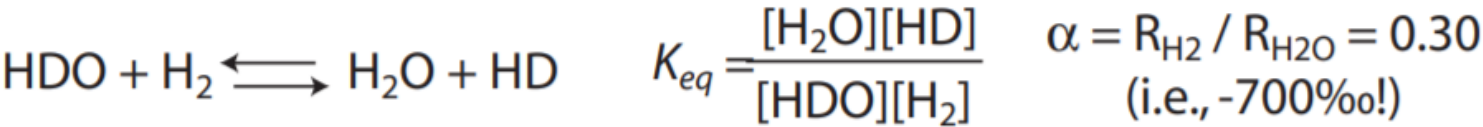
Wiederhold, Jan G., (2015) Environmental Science & Technology

Équilibre (processus bidirectionnel)

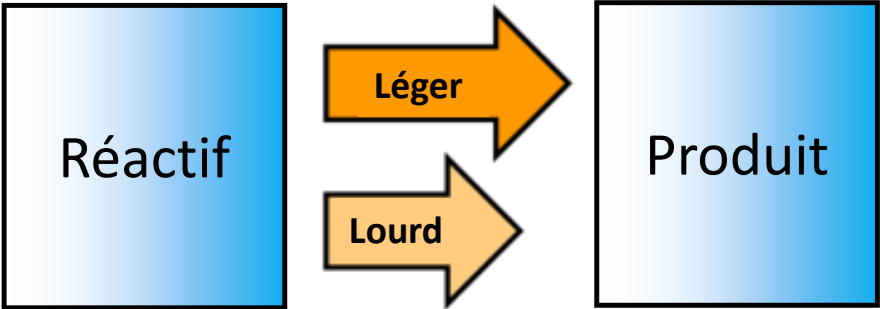
L'énergie de vibration, rotation et translation dépend des masses des atomes



- Le rapport des isotopes dans chaque compartiment est constant à une température donnée



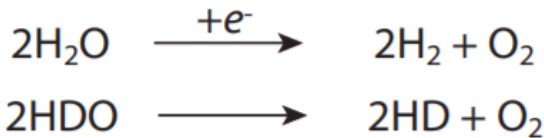
Cinétique (processus unidirectionnel, p.ex. enzyme)



Résulte du mouvement

$$E = \frac{1}{2}mv^2 \quad \frac{v_{\text{léger}}}{v_{\text{lourd}}} \propto \sqrt{\frac{m_{\text{lourd}}}{m_{\text{léger}}}}$$

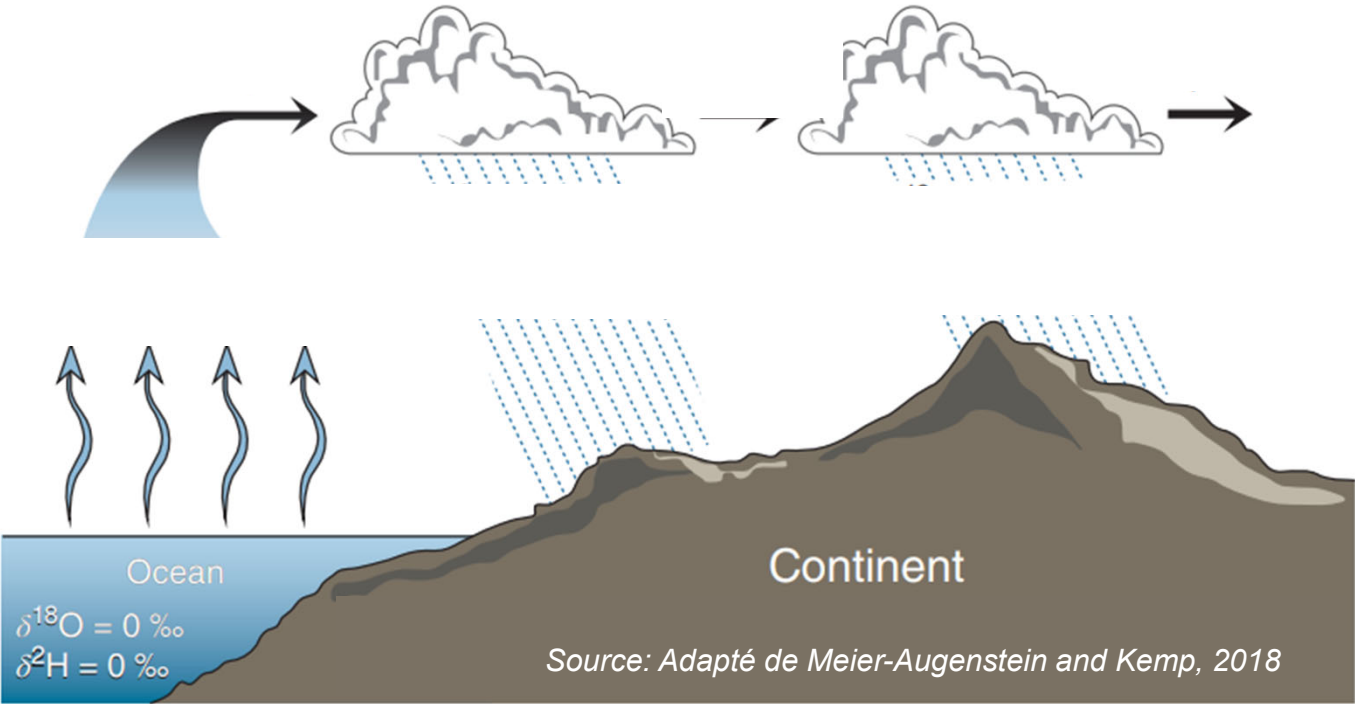
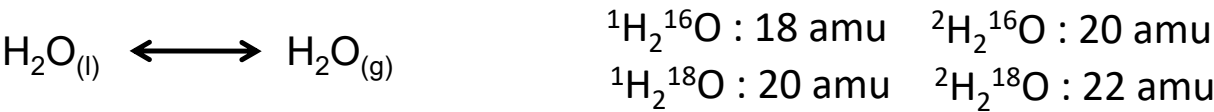
- La différence dans les **vitesse de réaction** des isotopologues dépend **du ratio des masses** des isotopes impliqués et de leur **fréquence de vibration**
- Processus **irréversible** (redox, enzymes)
- Plus le lien à une énergie faible, plus l'énergie potentielle pour le briser est faible



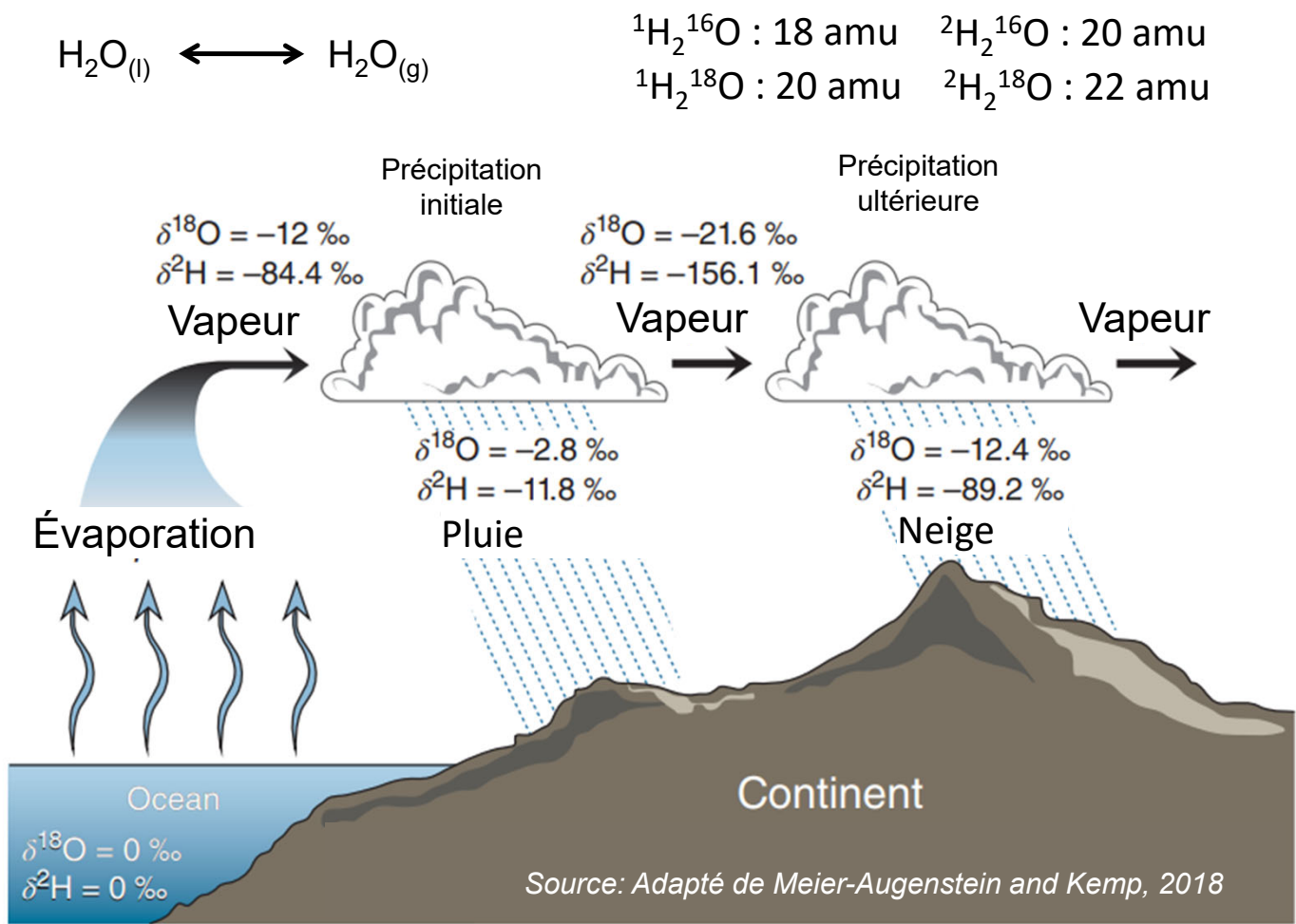
Fractionnement à l'équilibre et cinétique: résumé

- **Équilibre** : Le rapport des isotopes dans chaque compartiment est constant à une température donnée
- **Cinétique**: Les liens avec des isotopes légers sont brisés plus rapidement dans une réaction unidirectionnelle

Fractionnement isotopique: cycle de l'eau

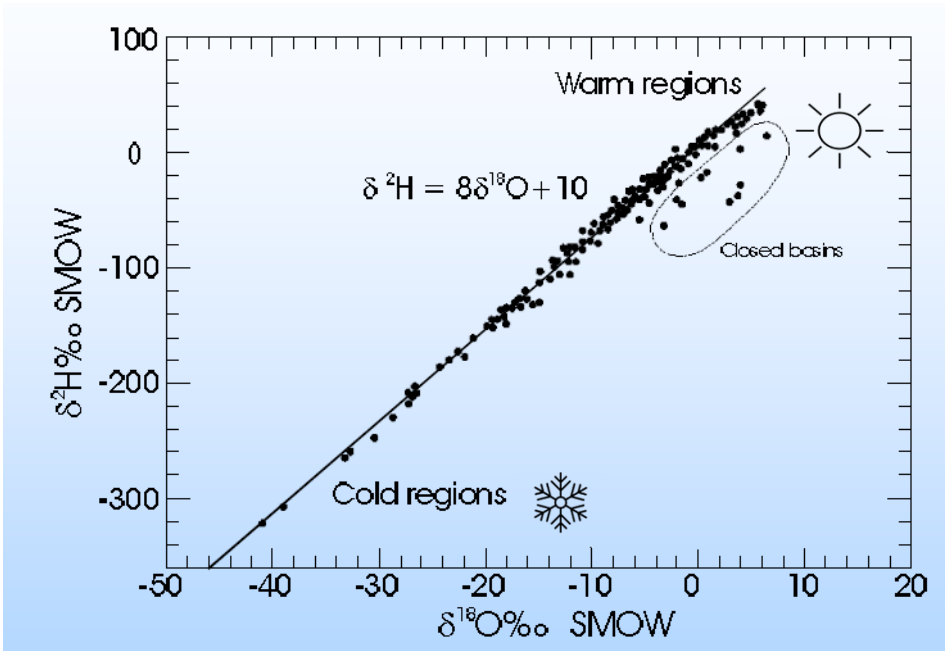
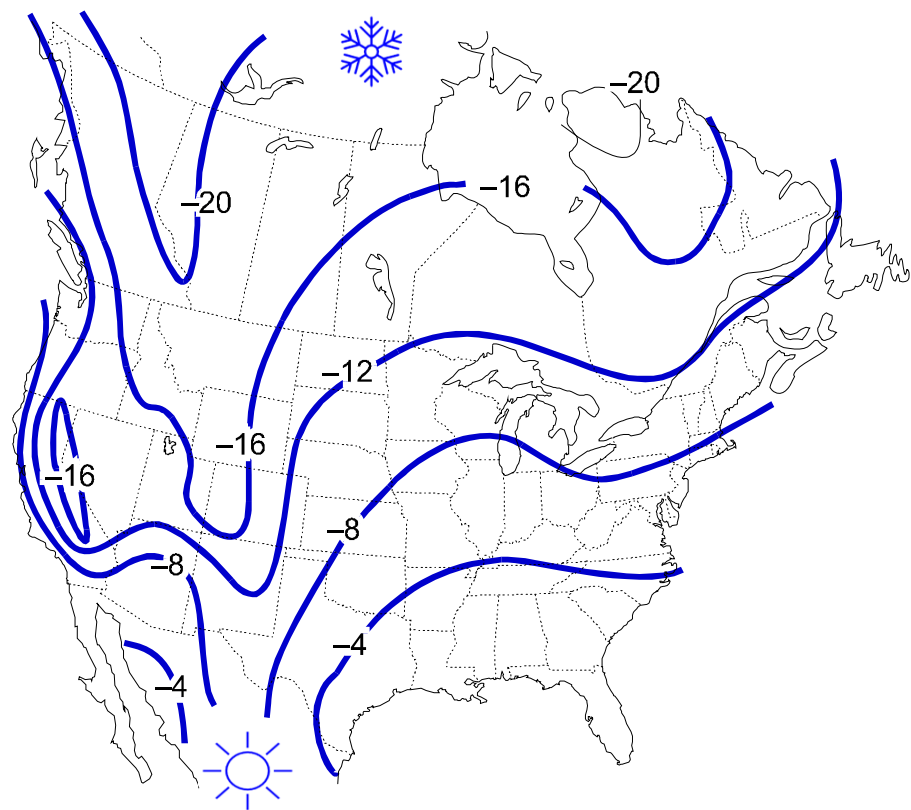


Fractionnement isotopique: cycle de l'eau



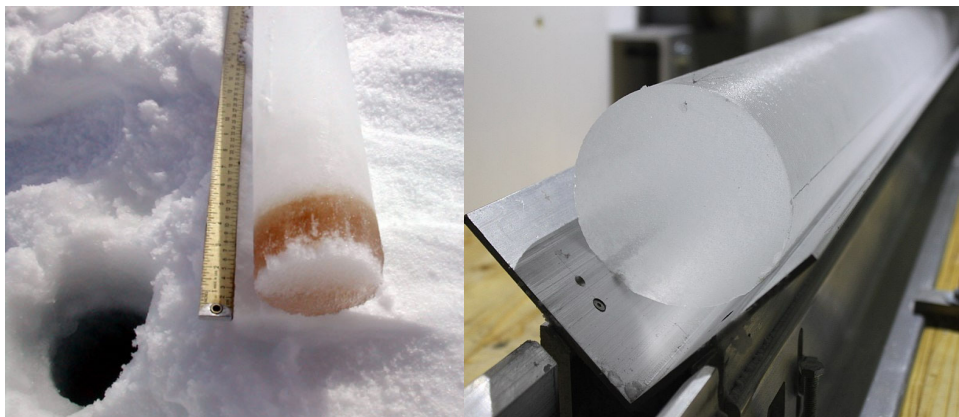
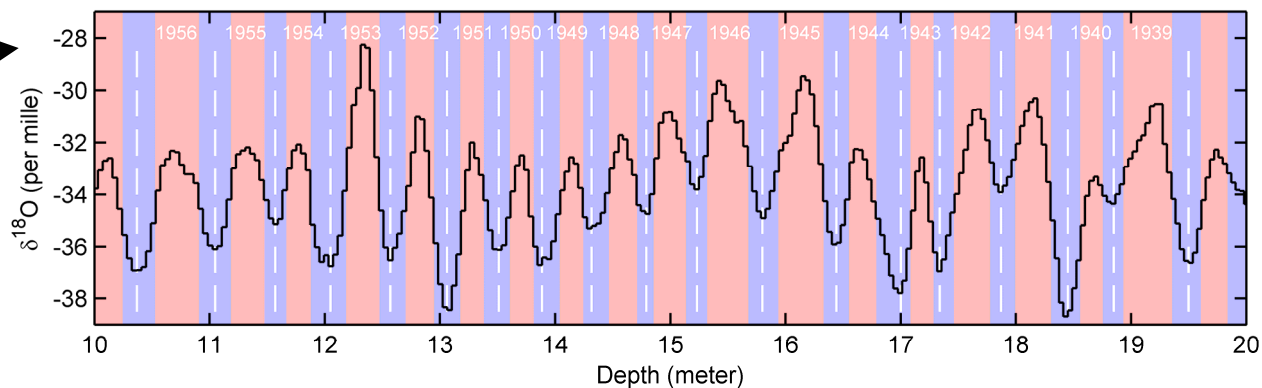
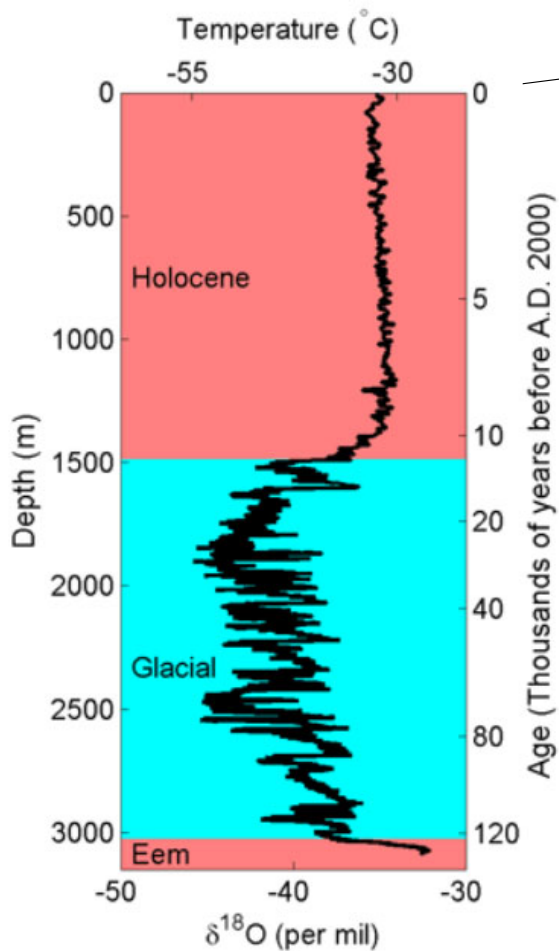
Fractionnement d'équilibre: la condensation

Distribution du d¹⁸O dans l'eau de pluie



Kohn, M. J., & Dettman, D. L. (2007); Al-Gamal, Samir (2020), 978-620-2-67836-0

Fractionnement à l'équilibre: archive de glace



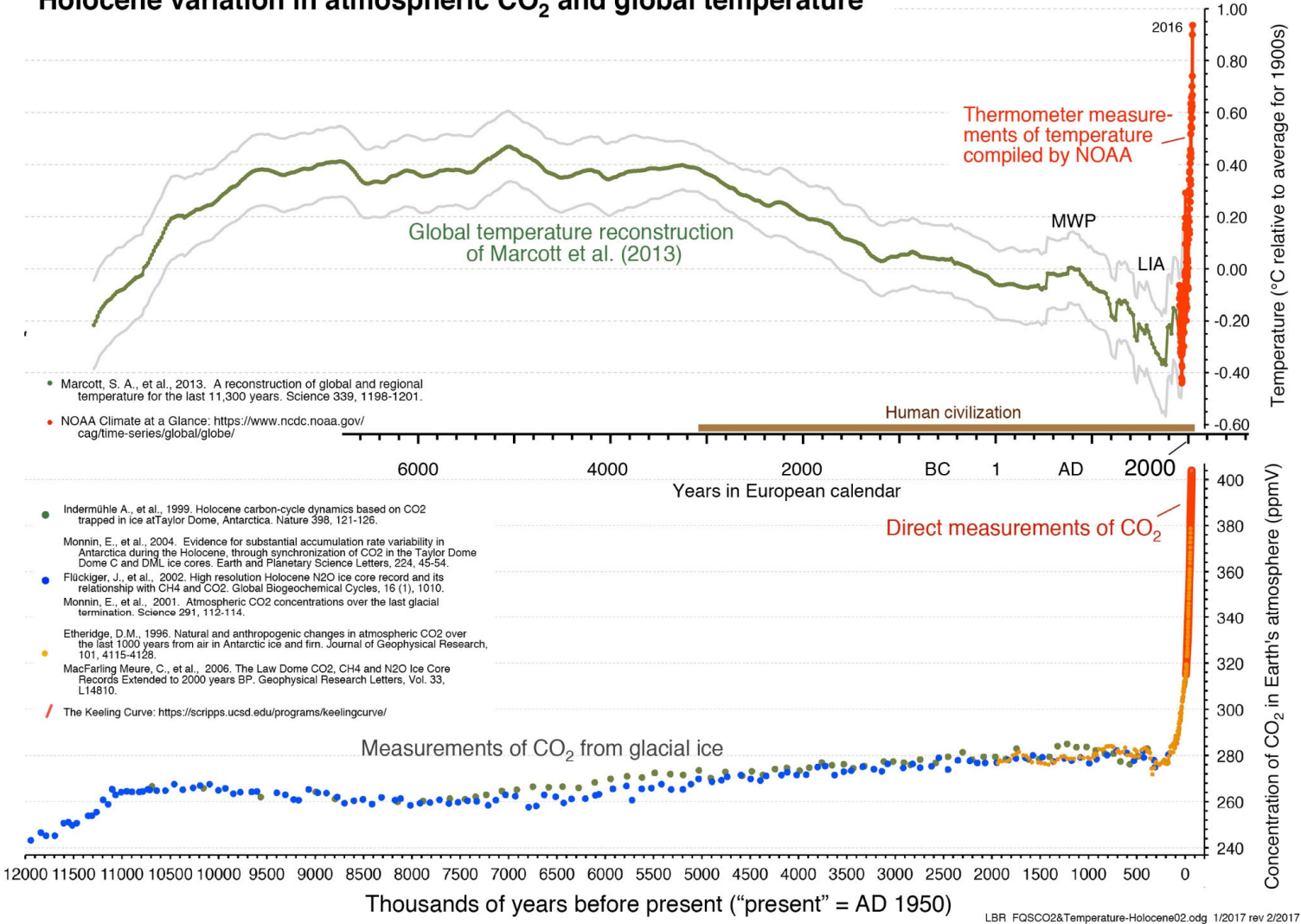
<https://www.iceandclimate.nbi.ku.dk/>

Les paléothermomètres montrent les températures de l'holocène

Railsback, L. B. (2017). Holocene variation in atmospheric CO₂ and global temperature. *Fundamentals of Quaternary Science*.

Isotopie · Cycle de l'eau

Holocene variation in atmospheric CO₂ and global temperature

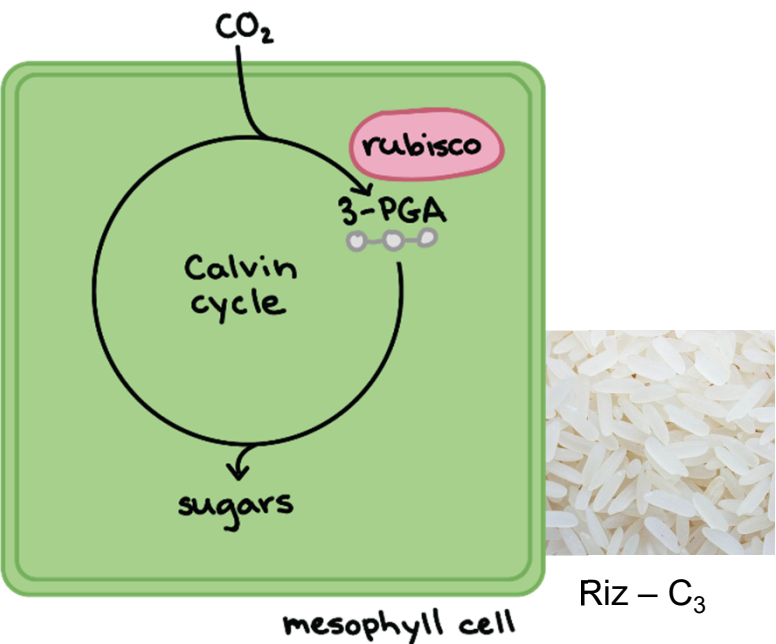


Fractionnement à l'équilibre et cinétique

A. Chemin C₃ (Calvin)

- Plus commun (95% plantes)

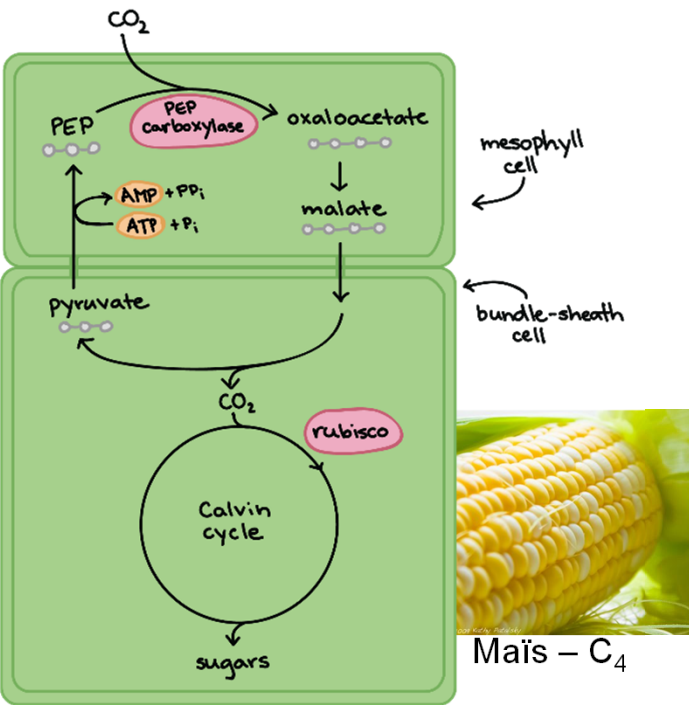
C₃ PATHWAY



B. Chemin C₄ (Hatch-Slack)

- Moins commun (5% plantes, 25% du C fixé)

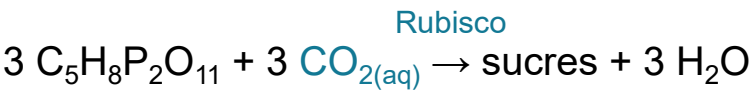
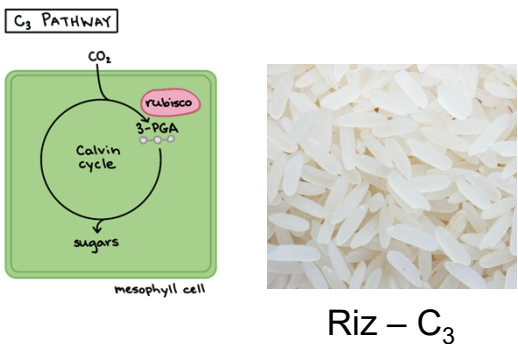
C₄ PATHWAY



Fractionnement à l'équilibre et cinétique

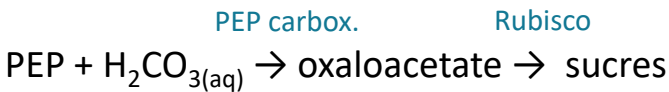
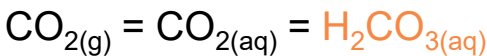
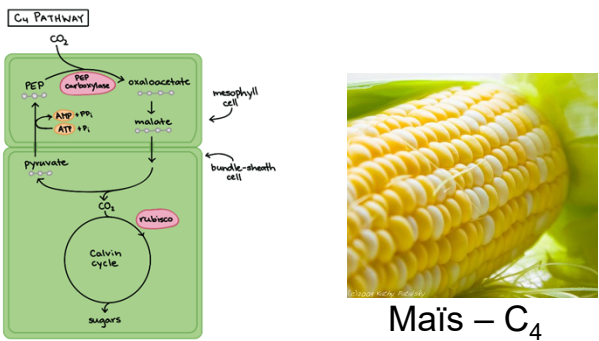
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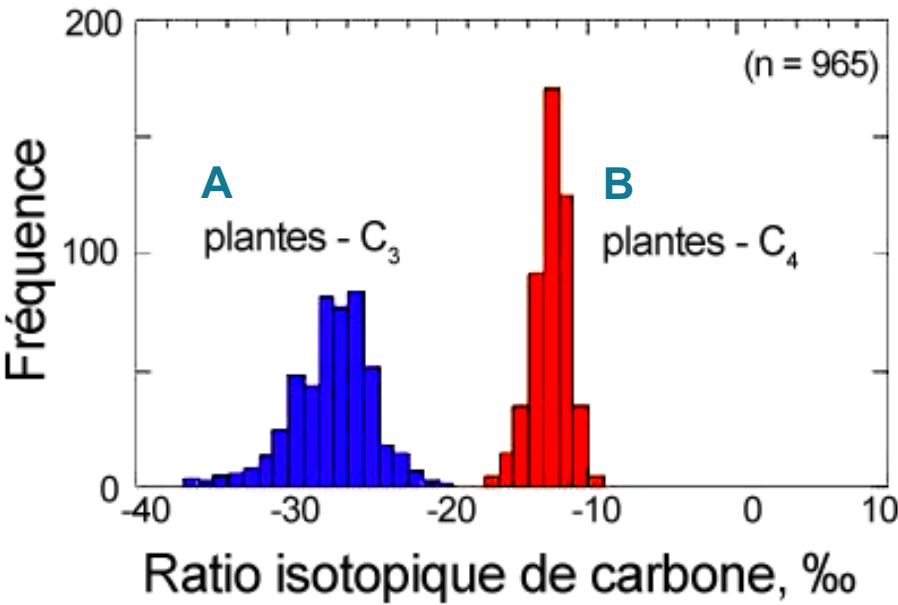


B. Chemin C₄ (Hatch-Slack)

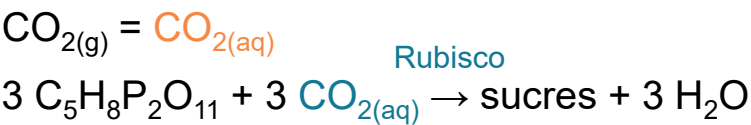
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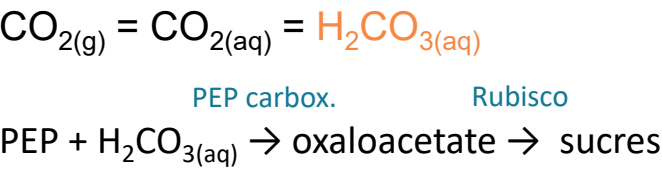
Fractionnement à l'équilibre et cinétique



A. Chemin C₃ (Calvin)



B. Chemin C₄ (Hatch-Slack)



$\delta^{13}\text{C}$ – Produits alimentaires

La fraude alimentaire est un fléau global (OCDE, 2007) d'une valeur de plus de 200 G\$ par année :

Fraude de contenu:

- Huile d'olive
- Huile de noix
- **Sirop d'érable**
- Médicaments
- Miel

Fraude de provenance:

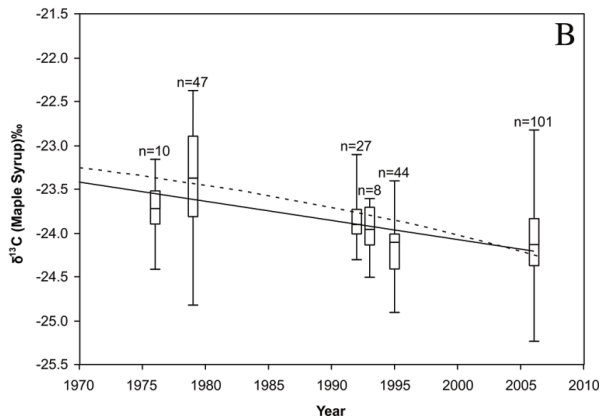
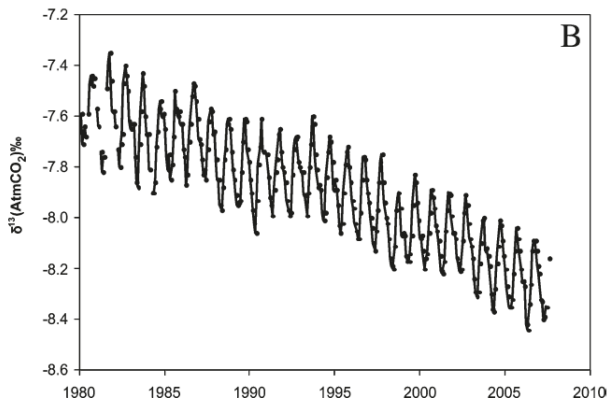
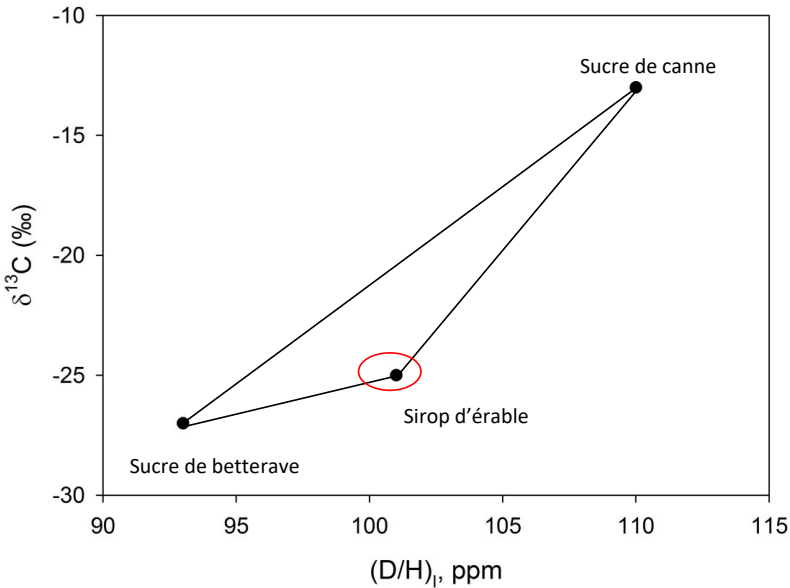
- Vins (français et italiens)
- Whisky (écossais)
- Fragrance et parfums (parfums de fruits)
- Textile (coton égyptien)



*Steeve Bourdeau, Fraudeur
(Enquête, Radio-Canada, Avril 2026)*

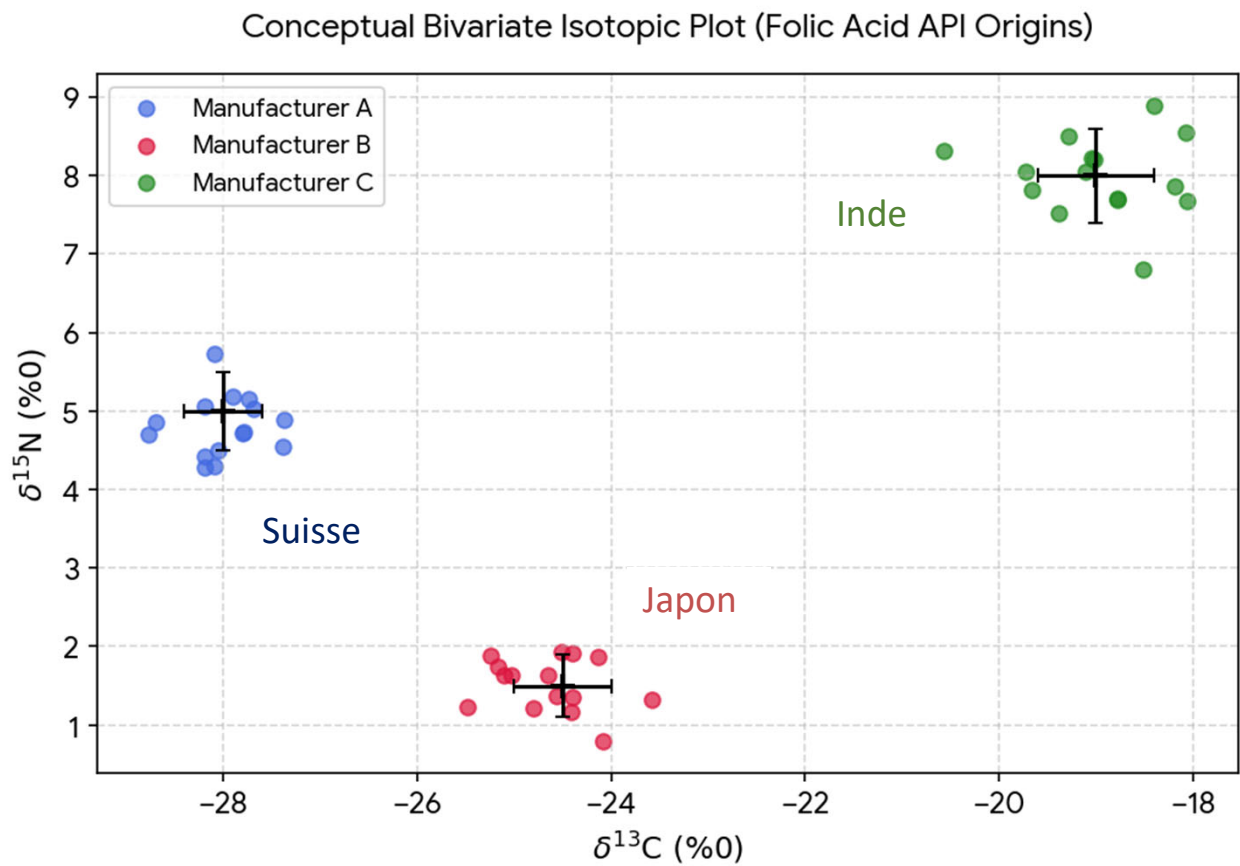
Attention aux marques génériques ! (Amazon, Temu, Shein = 💀)

$\delta^{13}\text{C}$ – Sirop d'érable

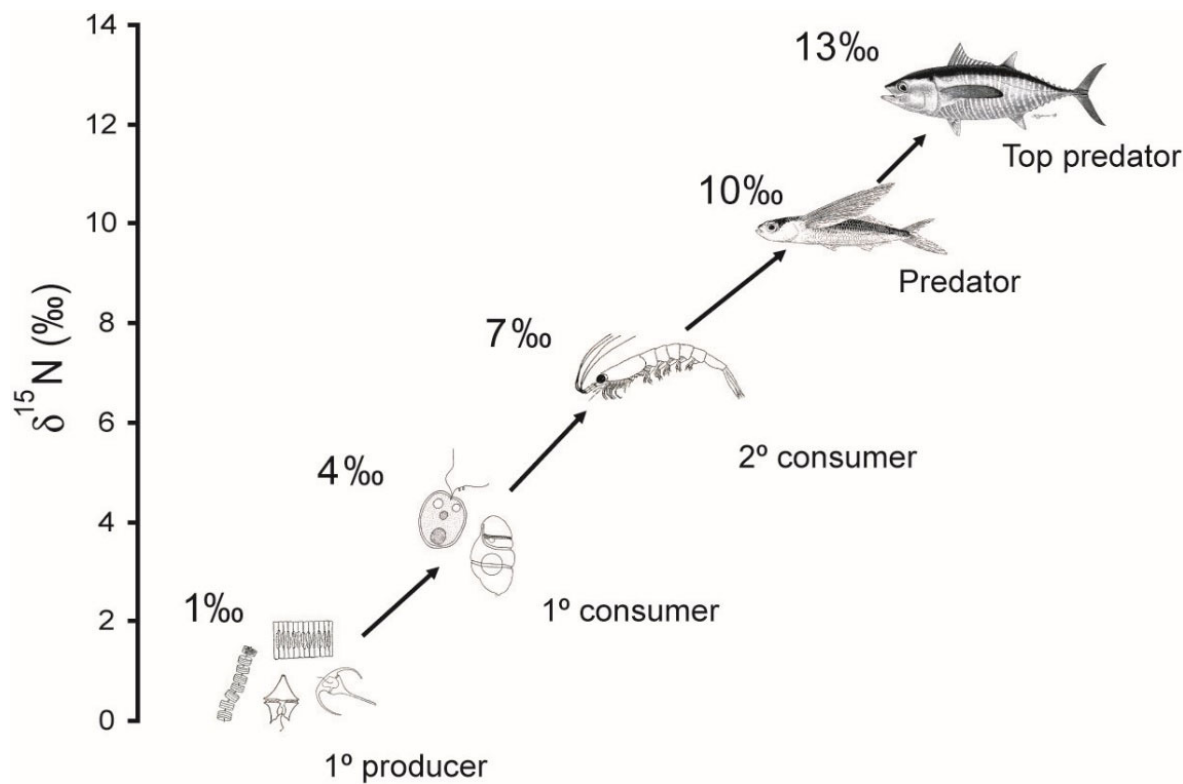


G.G. Martin et al., *J. Agric. Food Chem.* **1996**, 44, 3206-3213; WH Peck
SC Tubman, *J. Agric. Food Chem.* **2010**, 58, 2364-2367

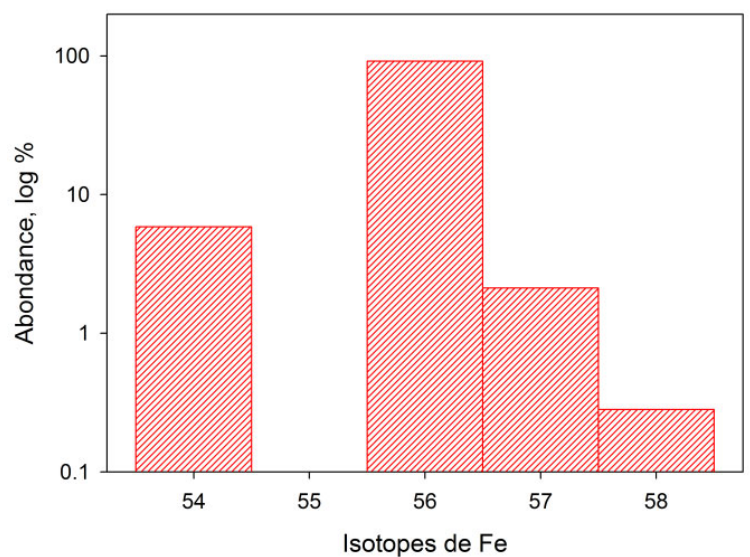
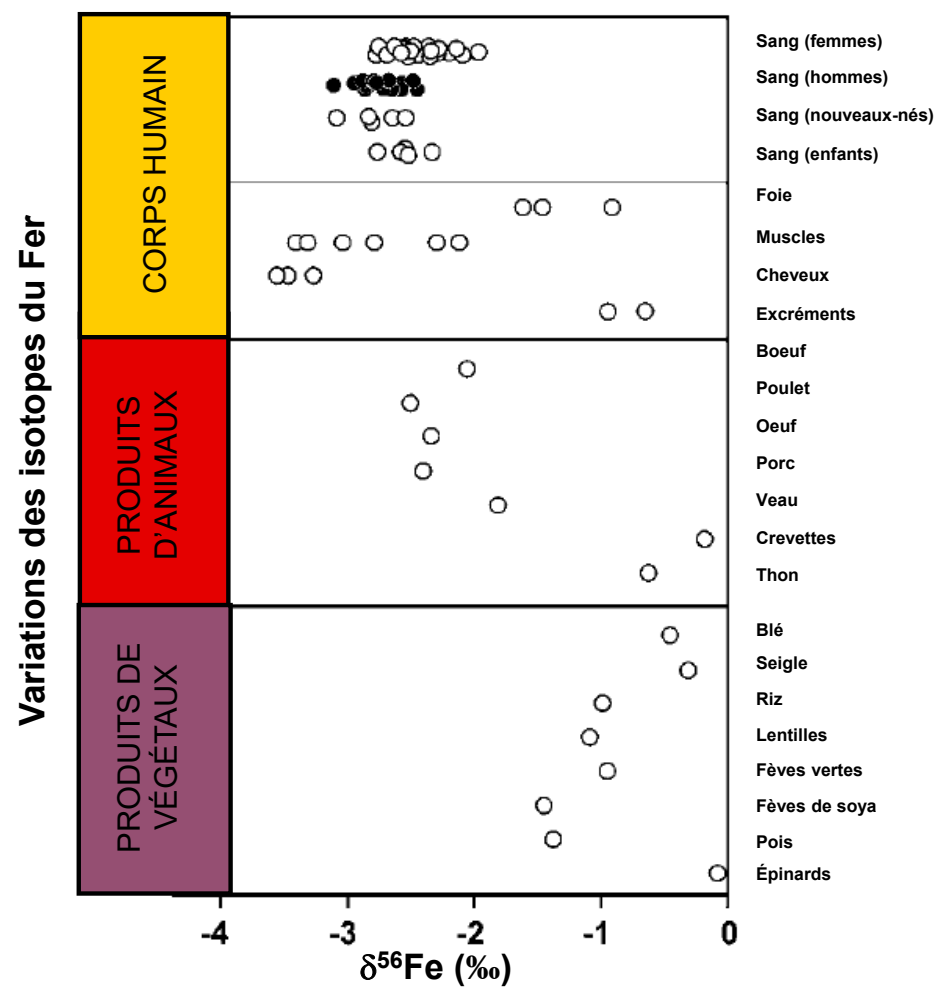
$\delta^{15}\text{N}$ – Folic Acid (“Booster” de Chimiothérapie)



$\delta^{15}\text{N}$ – Chaîne alimentaire (trophique)

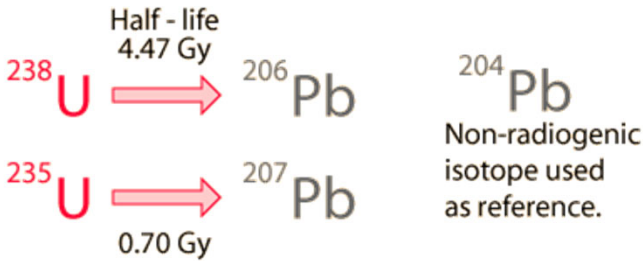


$\delta^{56}\text{Fe}$ – Métabolisme du fer

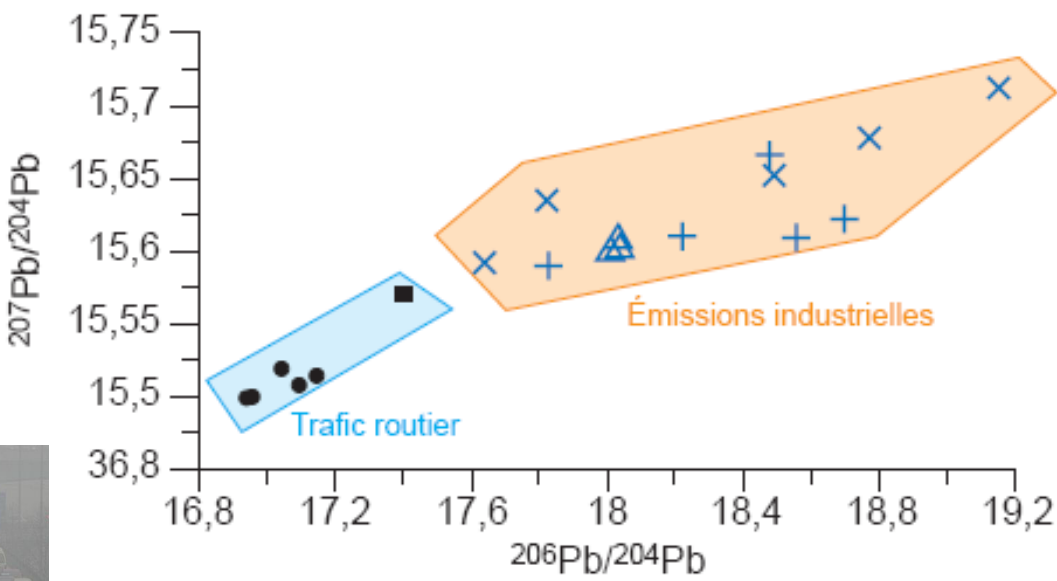


T. Walczyk et F. Von Blanckenburg, *Science*, 295 (2002) 2065

Discrimination des sources de Pb industrielles

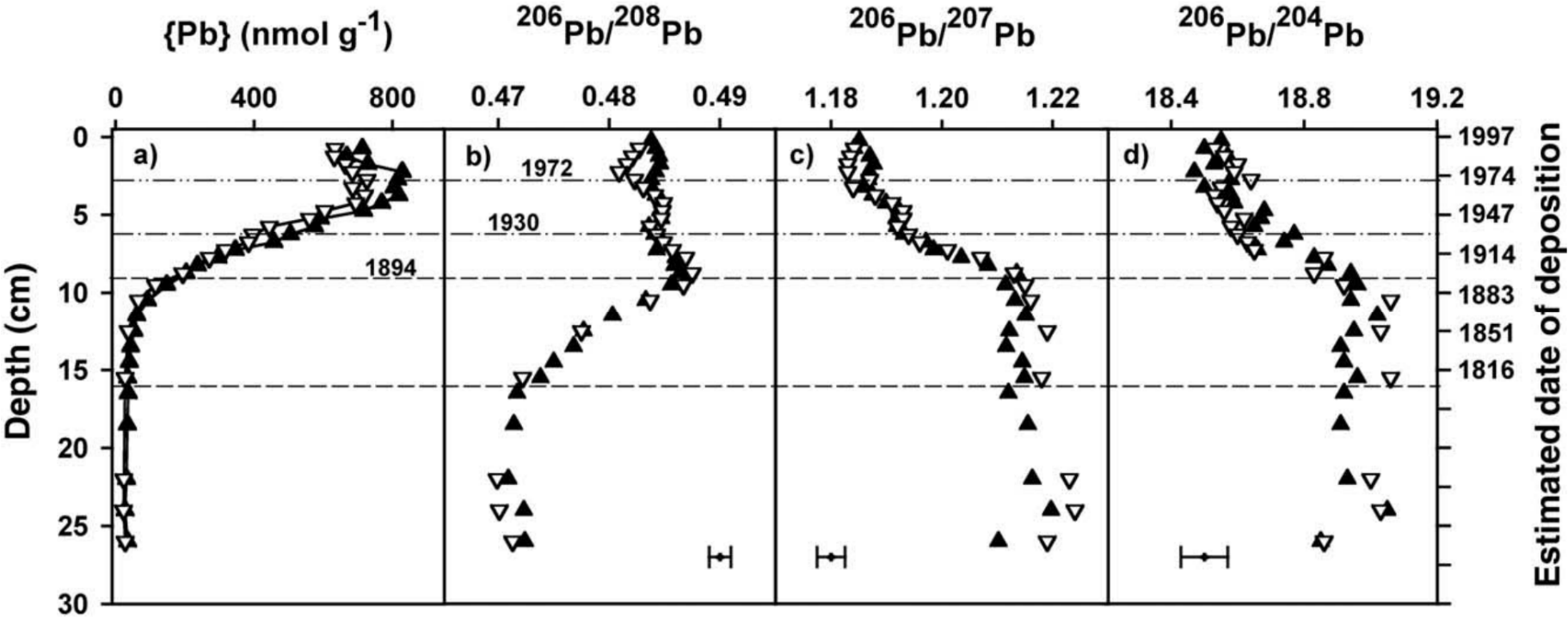


Ratio des isotopes stables du Pb – Sources de pollution



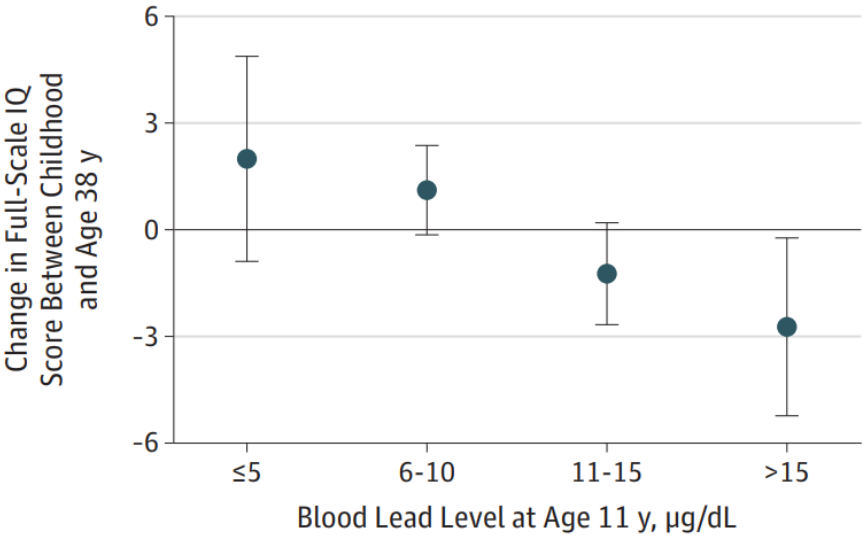
Compositions isotopiques du plomb des sources de pollution dans l'atmosphère parisienne.

Concentration et signature du Pb, région de Québec

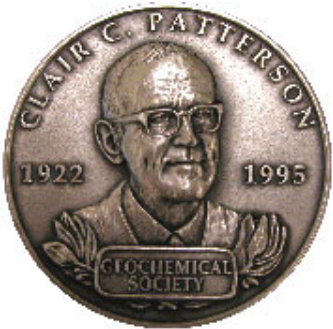


Déclin cognitif causé par le Pb

A Change in full-scale IQ by childhood blood lead levels



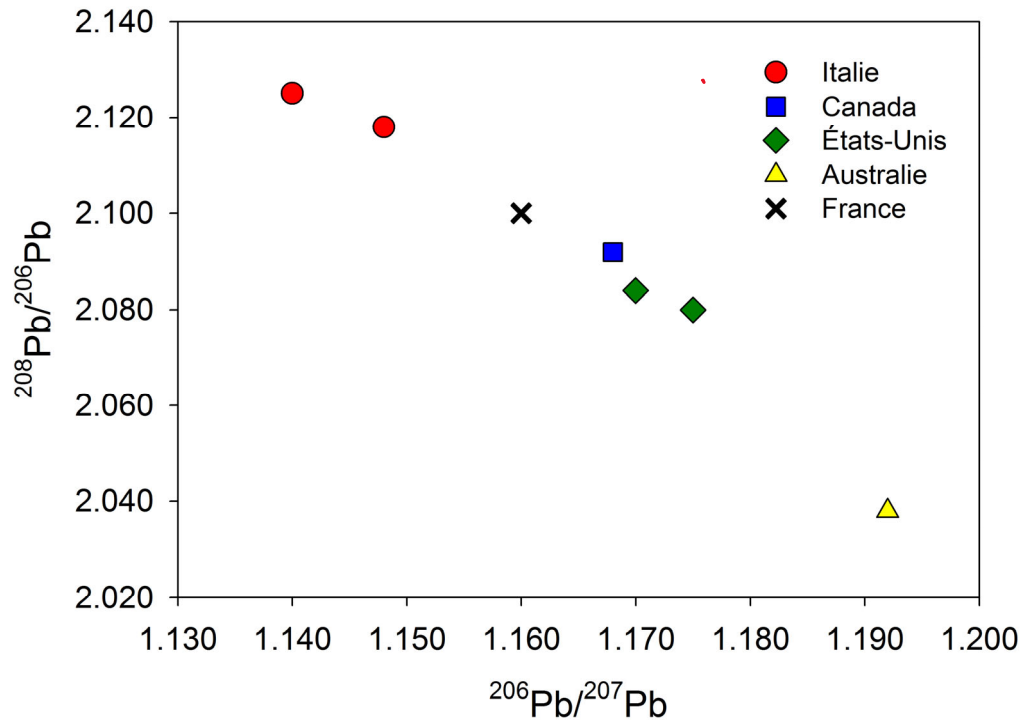
Reuben et al. JAMA. 2017



Clair Patterson Award

“Contaminated and Natural Lead Environments of Man”
1965

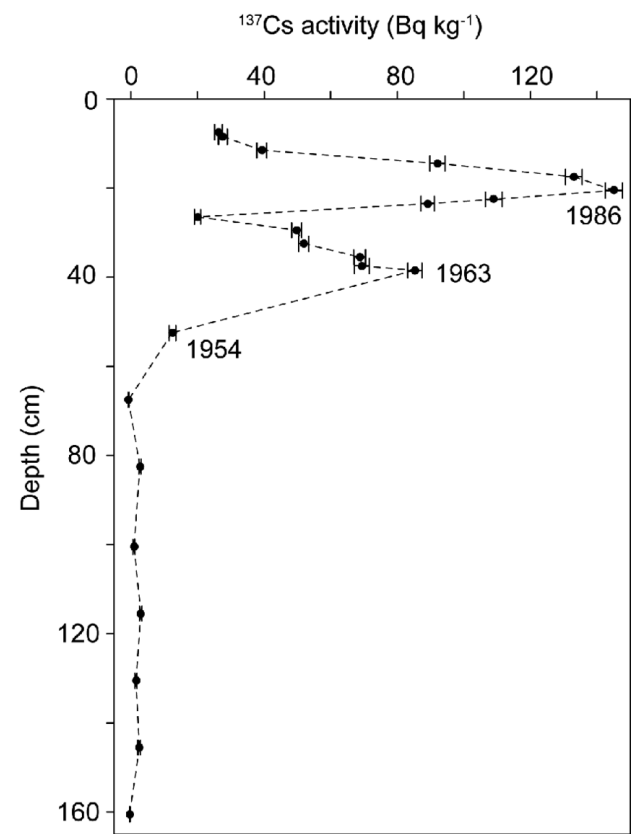
Ratio Pb – Contrefaçon du vin



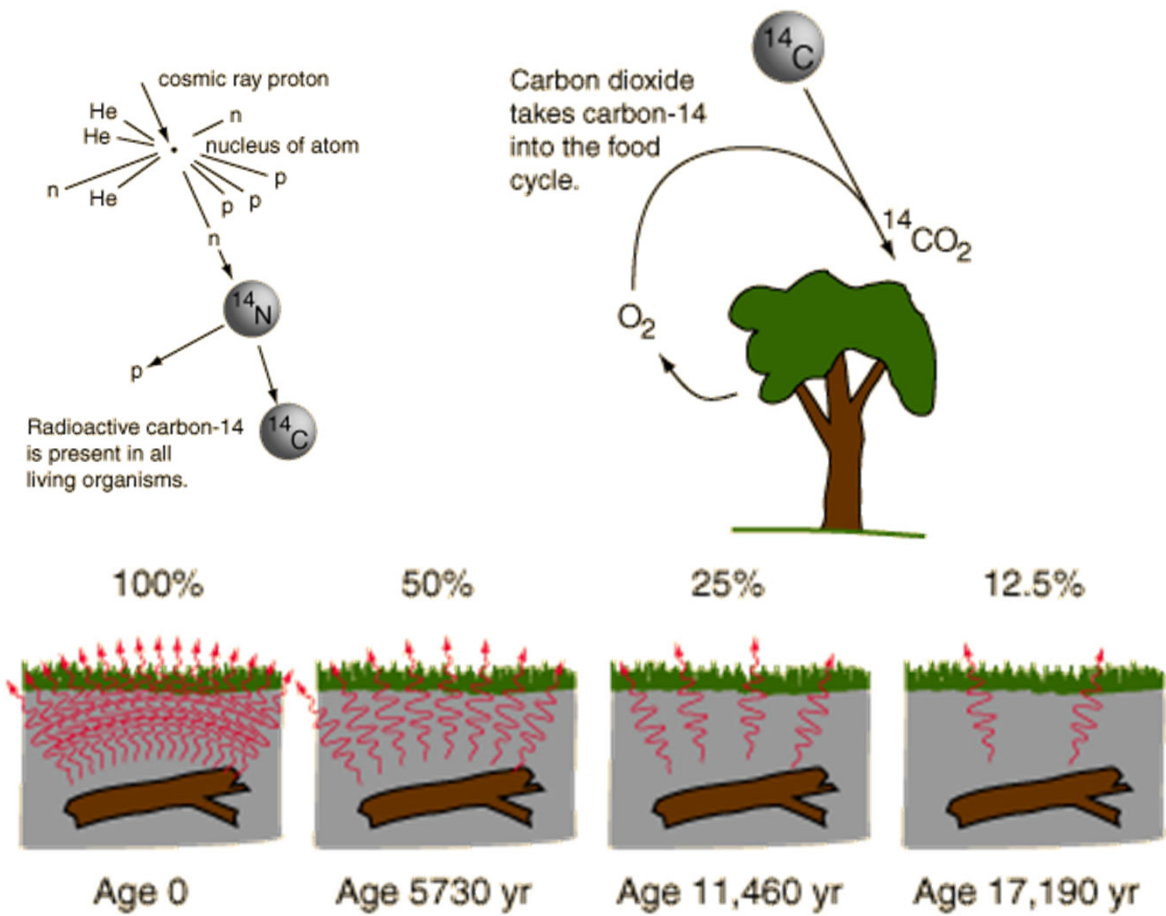
Ratios isotopiques du plomb dans le vin rouge de 5 pays.

Tian, X et al., J. Anal. At. Spectrom., 15 (2000) 829.

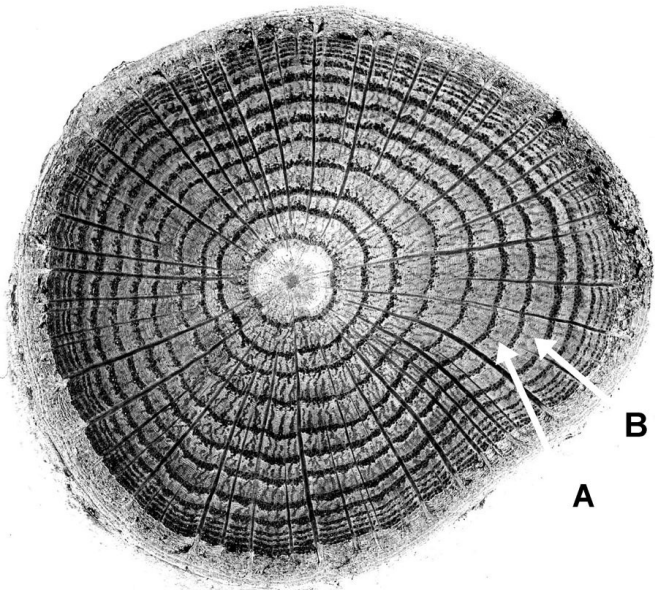
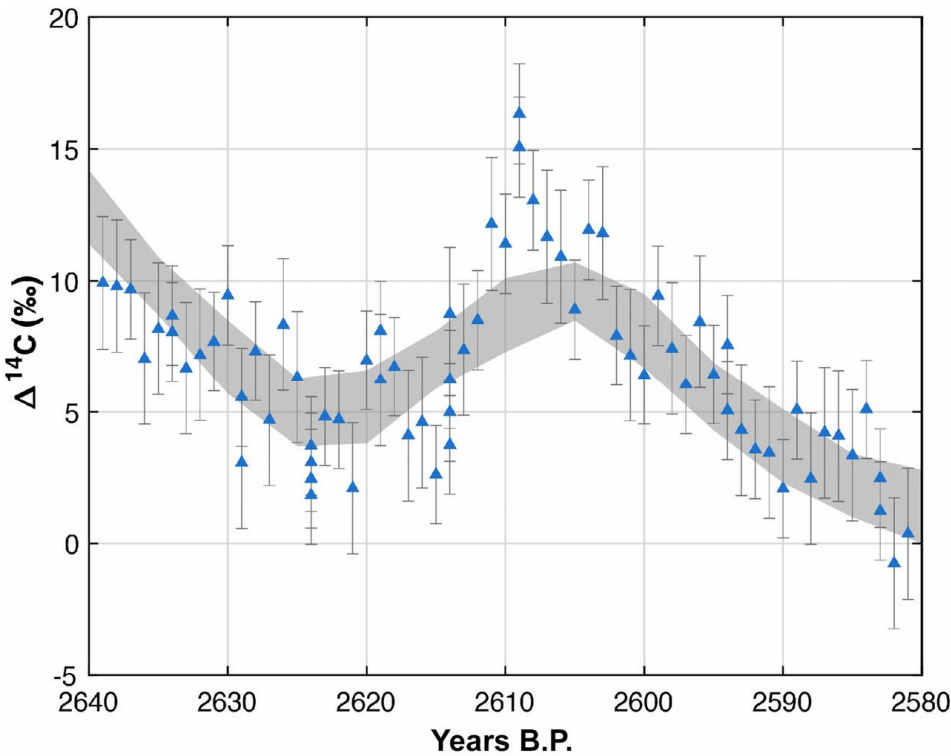
Suivi environnemental : radiocésium (^{137}Cs)



Datation au ¹⁴C



Reconstruction du flux de neutrons solaire



Applications pratiques des isotopes

La signature isotopique est un indice:

Des processus (bio)chimiques (isotopes **légers**)

Des sources (isotopes **lourds**)

De l'âge (isotopes **radioactifs**)

H & O : météorologie

O: paléothermométrie

C,N : niveau trophique

Pb: enquêtes environnementales

Géochronologie:

^{210}Pb : < 150 a

^{14}C : < 60 000 a

U/Th: < 1 million a

Pb/Pb: < 3 milliards a